Kernix 0.8.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

21 workloads

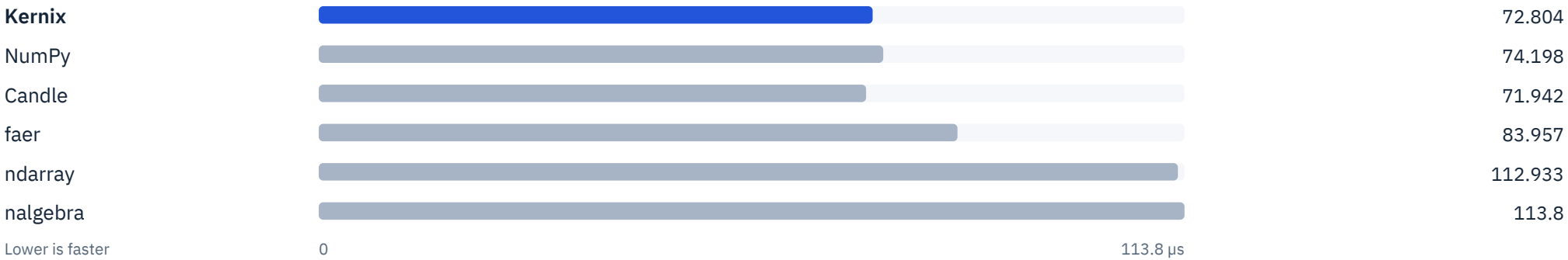
88 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

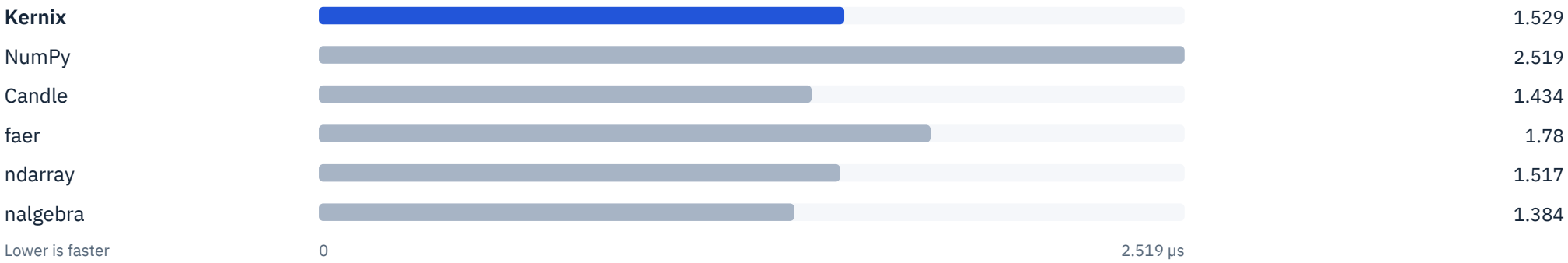
Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

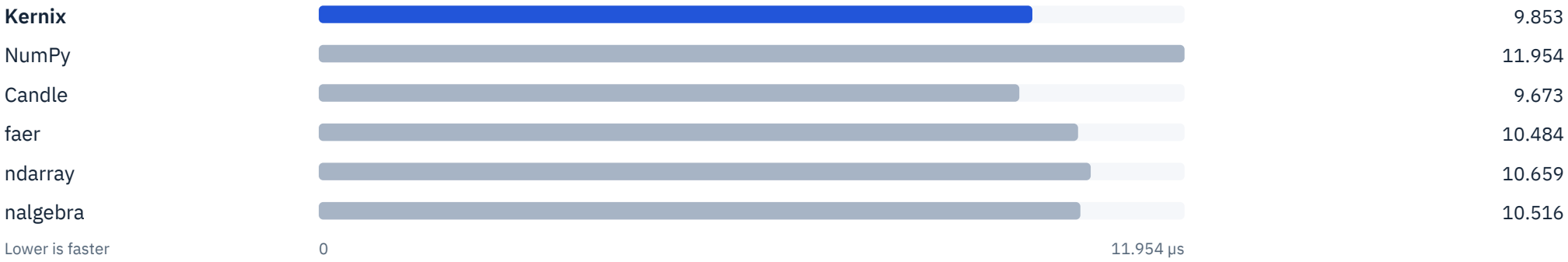
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



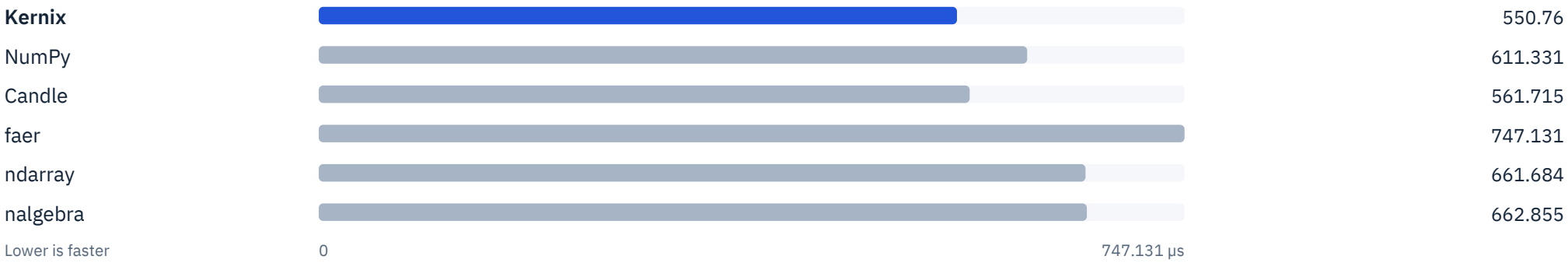
Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

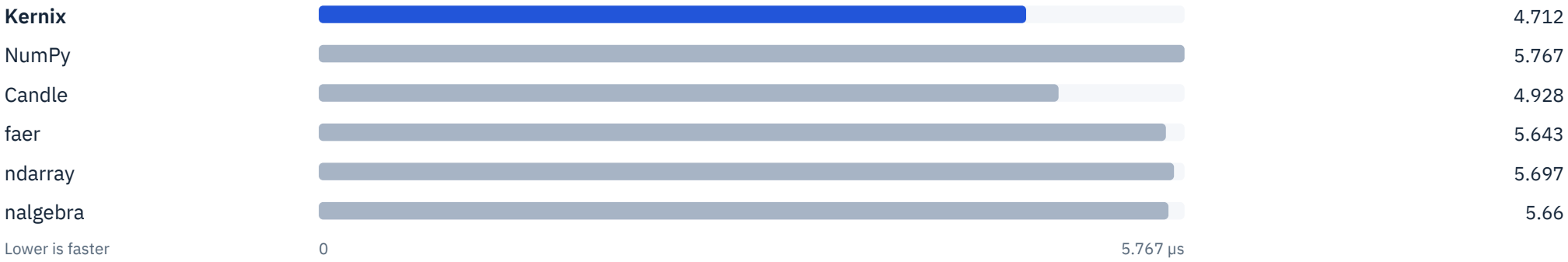
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



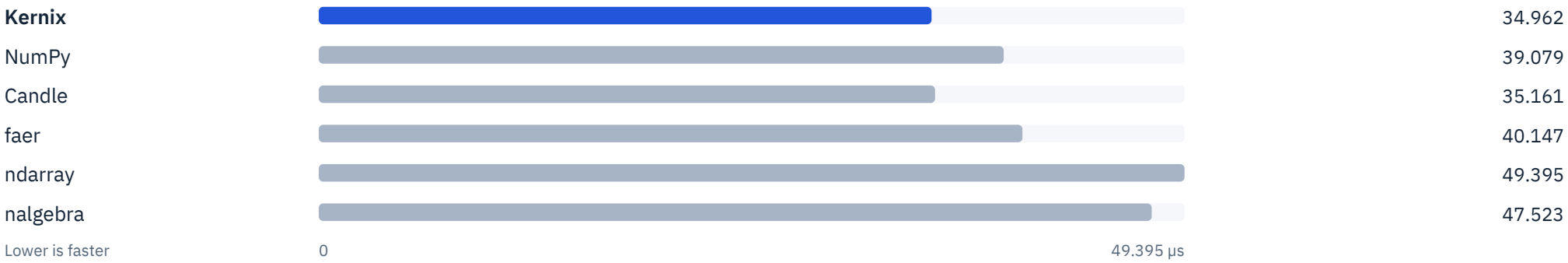
Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

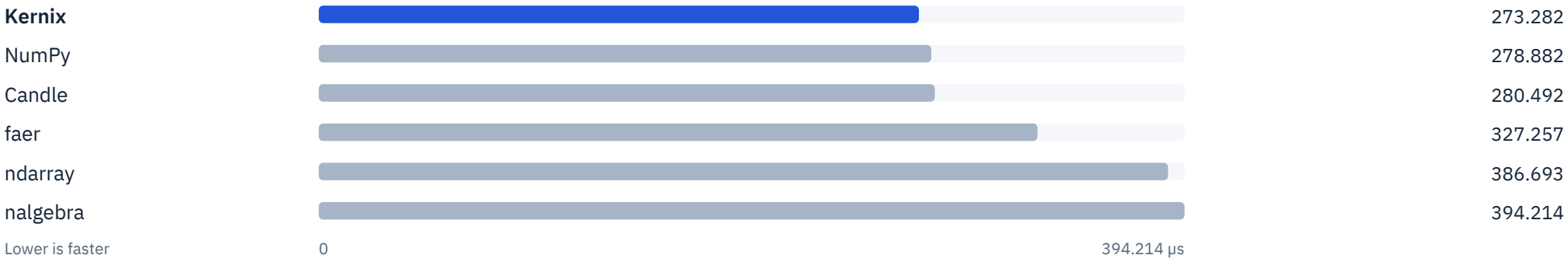
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.8.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches

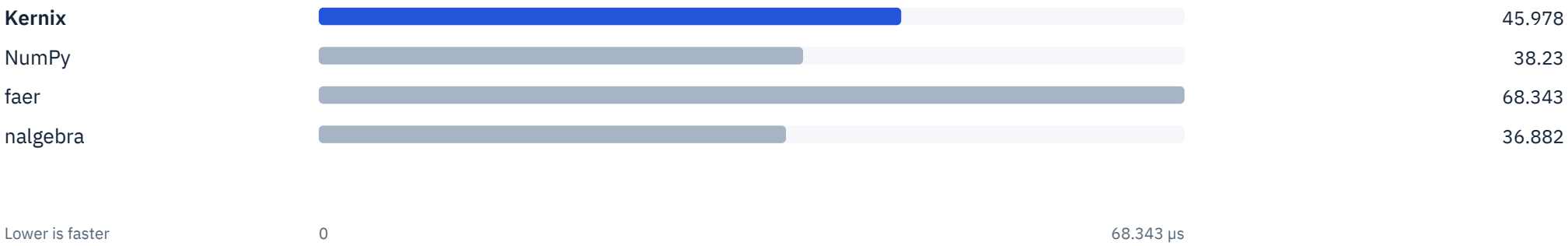


Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



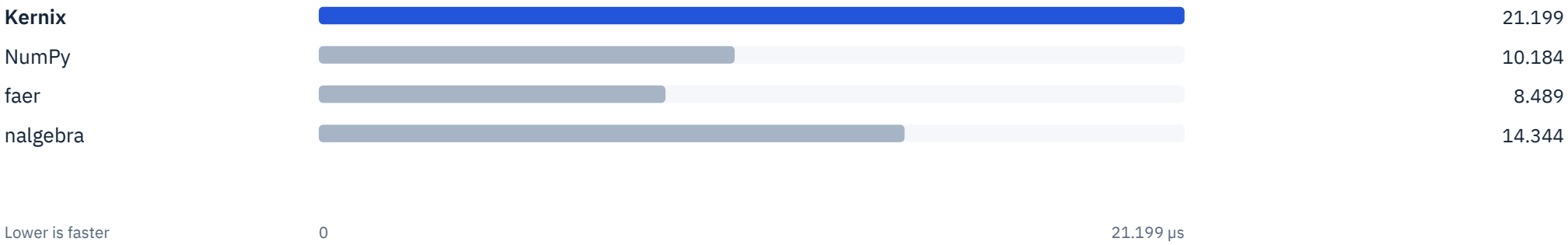
Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

Per-call median

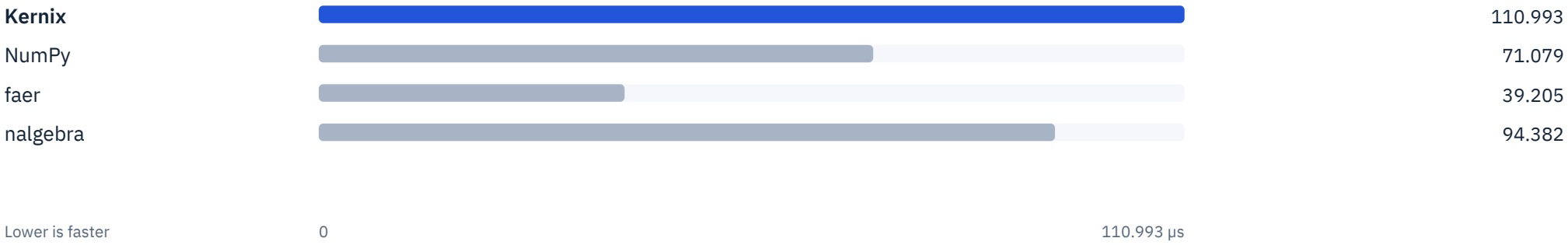
64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

128 × 128 · f64 · 1 worker · allocated output

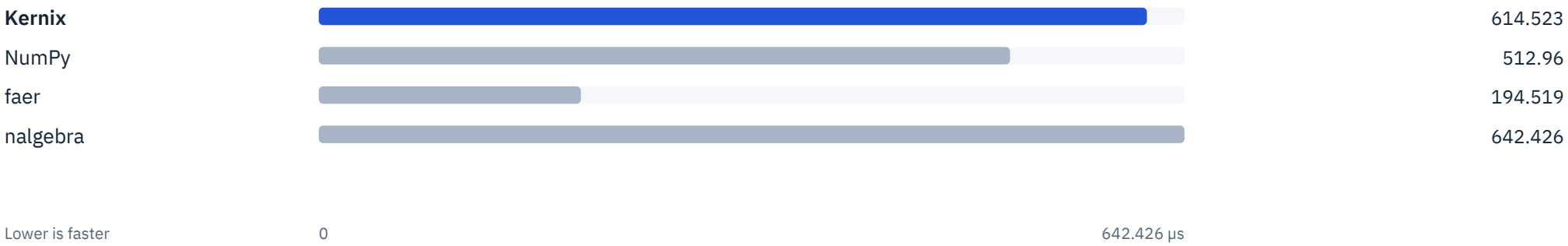


Kernix 0.8.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

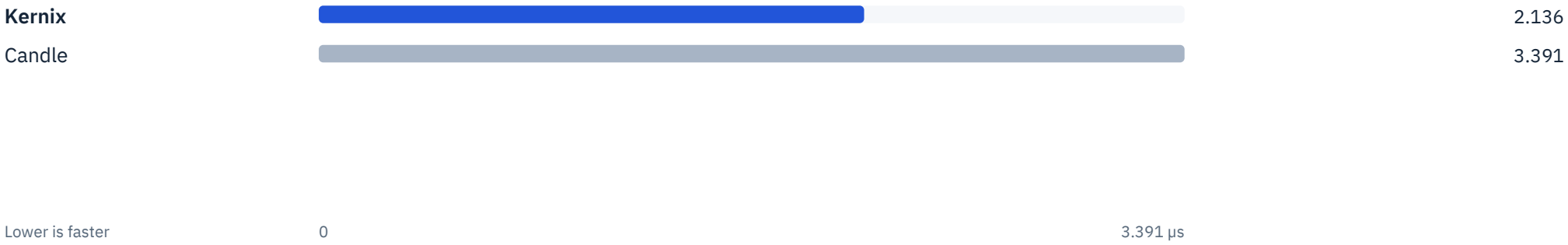
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

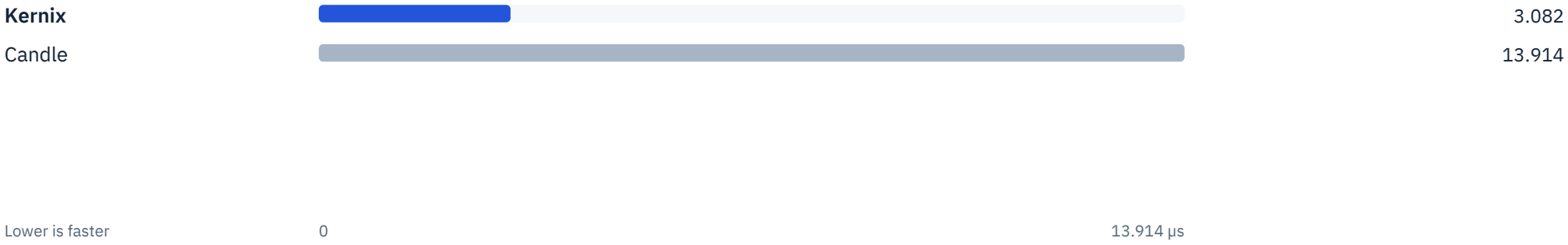
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.8.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented

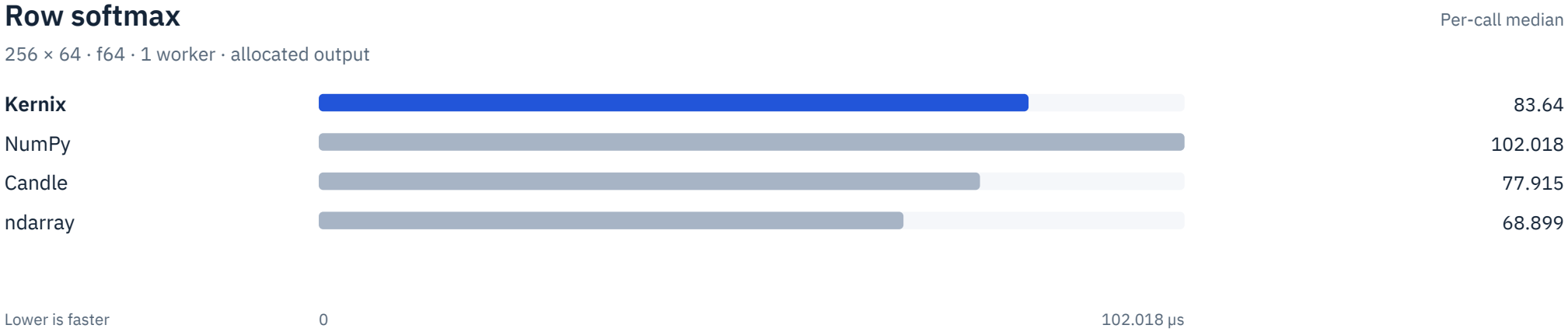


Kernix 0.8.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



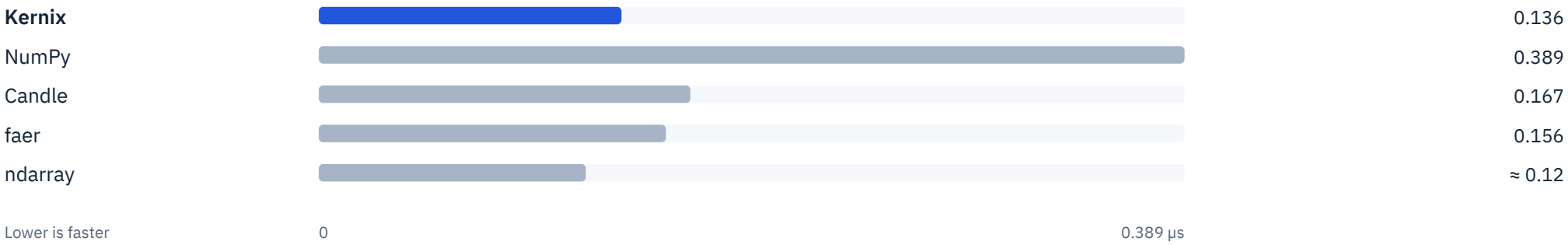
Kernix 0.8.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1529	1.529
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1517	1.517
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1780	1.78
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1434	1.434
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2519	2.519
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1384	1.384
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	10516	10.516
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	11954	11.954
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	9853	9.853
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	10659	10.659

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	10484	10.484
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	9673	9.673
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	113800	113.8
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	83957	83.957
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	112933	112.933
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	72804	72.804
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	74198	74.198
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	71942	71.942
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	662855	662.855
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	747131	747.131

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	611331	611.331
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	550760	550.76
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	561715	561.715
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	661684	661.684
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	5660	5.66
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	4928	4.928
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	4712	4.712
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	5767	5.767
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	5643	5.643
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	5697	5.697

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	49395	49.395
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	39079	39.079
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	47523	47.523
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	34962	34.962
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	35161	35.161
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	40147	40.147
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	327257	327.257
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	280492	280.492
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	394214	394.214
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	273282	273.282

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	278882	278.882
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	386693	386.693
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	2153	2.153
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2945	2.945
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2018	2.018
Thin QR / Kernix 96 × 32 · f64	45978	45.978
Thin QR / faer 96 × 32 · f64	68343	68.343
Thin QR / nalgebra 96 × 32 · f64	36882	36.882
Thin QR / NumPy 96 × 32 · f64	38230	38.23
Thin QR / Kernix 256 × 64 · f64	480702	480.702

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	399014	399.014
Thin QR / nalgebra 256 × 64 · f64	368639	368.639
Thin QR / NumPy 256 × 64 · f64	336724	336.724
Cholesky factorization / nalgebra 64 × 64 · f64	14344	14.344
Cholesky factorization / Kernix 64 × 64 · f64	21199	21.199
Cholesky factorization / faer 64 × 64 · f64	8489	8.489
Cholesky factorization / NumPy 64 × 64 · f64	10184	10.184
Cholesky factorization / NumPy 128 × 128 · f64	71079	71.079
Cholesky factorization / nalgebra 128 × 128 · f64	94382	94.382
Cholesky factorization / faer 128 × 128 · f64	39205	39.205

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	110993	110.993
Cholesky factorization / faer 256 × 256 · f64	194519	194.519
Cholesky factorization / nalgebra 256 × 256 · f64	642426	642.426
Cholesky factorization / Kernix 256 × 256 · f64	614523	614.523
Cholesky factorization / NumPy 256 × 256 · f64	512960	512.96
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	2592	2.592
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	3958	3.958
Model inference / Candle batched normalized/sliced MLP · f64	3391	3.391
Model inference / Kernix batched normalized/sliced MLP · f64	2136	2.136
Model recording + forward / Kernix batched normalized/sliced MLP · f64	2582	2.582

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	5017	5.017
Model backward / Candle prepared graph → fresh gradients · f64	13914	13.914
Model backward / Kernix prepared graph → fresh gradients · f64	3082	3.082
Parameter update / Kernix transactional SGD, fixed gradients · f64	498.00000000000006	0.49800000000000006
Row softmax / NumPy 256 × 64 · f64	102018	102.018
Row softmax / Candle 256 × 64 · f64	77915	77.915
Row softmax / Kernix 256 × 64 · f64	83640	83.64
Row softmax / ndarray 256 × 64 · f64	68899	68.899
Create zeros / Kernix 64 × 64 · f32	136	0.136
Create zeros / faer 64 × 64 · f32	156	0.156

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / ndarray 64 × 64 · f32	119.99999999999999	0.11999999999999999
Create zeros / NumPy 64 × 64 · f32	389	0.389
Create zeros / Candle 64 × 64 · f32	167	0.167
Create ones / faer 64 × 64 · f32	268	0.268
Create ones / Kernix 64 × 64 · f32	203	0.203
Create ones / Candle 64 × 64 · f32	226	0.226
Create ones / NumPy 64 × 64 · f32	1336	1.336
Create ones / ndarray 64 × 64 · f32	140	0.14